Checkpoint 4 - Genetic algorithm with repair (role GA-A)

Diar Telgarin, group IT-2401, team T10 · partner GA-P: Inkar Abdimazhit · instance *cp34_instance(10, "B", 2)* (S = 11 servers, M = 50 VMs, zones) · z_{LP} = 8.3395, z_{ref} = 8.3828 · budget 100 000 evaluations, seeds 0-9.

Method. Individual = server of each VM. Rule parameters (not tuned): N = 30, p = 0.9 (uniform crossover), 1-2 mutated c components (a VM moved to another allowed server), tournament of size 2, elitism 1. Each offspring is repaired *before* it is evaluated: VMs on a forbidden zone are moved to an allowed server, then from every overloaded server the smallest VM that cures the overload (else the largest) goes to the allowed server with room and the lowest CPU utilisation. The repair reads only loads and capacities (free); every offspring and every initial individual costs 1 evaluation, the elite copy is free, the last generation is cut short (exactly 100 000). Fitness = cost. My part: *ga.py* (loop, solve), the tests, the 10 final runs, the random-search baseline and the experiments; GA-P: *ga_problem.py*.

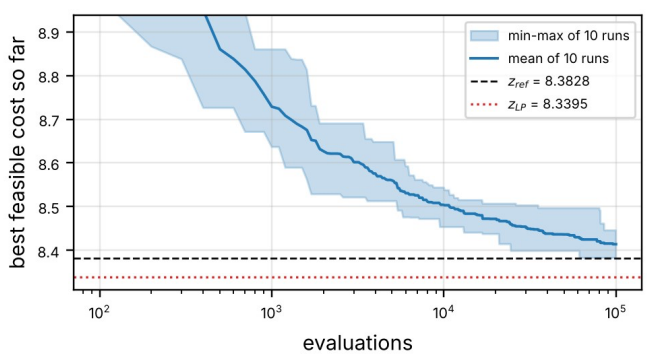
1. Prediction (written before the final runs)

Feasibility: 10 of 10 runs feasible - the repair always returns a feasible offspring and the fill ratio is low (CPU 233/504 = 46.2 %, memory 871/1725 = 50.5 %; only 7 premium VMs have a zone condition, each with 3-4 allowed servers). **Mean gap_ref :** between 0.2 % and 1.5 %, because z_{LP} and z_{ref} are only ~0.5 % apart and the convex latency has many nearly equal balanced assignments; the rest would come from the repair bias and from lost diversity, not from the budget. **Diversity:** the number of *distinct* individuals will not collapse (every offspring is mutated, exact clones are rare) and stays above ~10, but the genotypes will cluster around the elite: mean pairwise Hamming distance below ~5 of 50 within the first 1020 % of the budget (premature convergence in structure); a larger N will not improve the cost much.

2. Results

Table 1. Final runs, seeds 0–9 (results/Telgarin_runs.csv; baseline results/Telgarin_random.csv, same budget and seeds; std with divisor n - 1; time measured while another job was running).

	best	median	mean	worst	std	feasible	time	gap % (best / med / mean / worst)	gap % (best / med / mean / worst)
								$_{ref}$	$_{LP}$
GA, repair (N = 30)	8.3828	8.4123	8.4143	8.4459	0.0184	10/10	24.0 s	0.00 / 0.35 / 0.38 / 0.75	0.52 / 0.87 / 0.89 / 1.26
Random search	8.7888	8.8353	8.8468	8.9388	0.0541	10/10	64.3 s	4.84 / 5.40 / 5.53 / 6.63	5.11 / 5.61 / 5.73 / 6.70



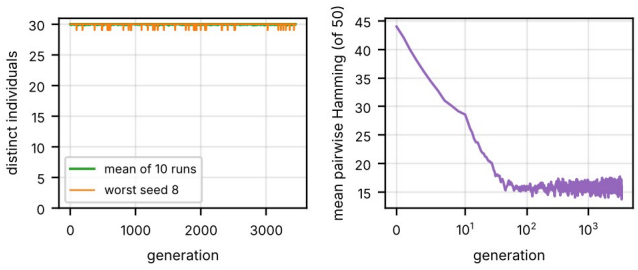


Figure F1 (left): best feasible cost against evaluations, mean of the 10 runs with the min–max band, z_{ref} and z_{LP} as lines (log x axis). **Figure F2** (right): diversity of the population (N = 30) against the generation: distinct individuals (mean of 10 runs and worst seed) and mean pairwise Hamming distance.

Table 2: worst seed	value	Effect of N (seeds 0–9, results/Telgarin_Nexp.csv)	N = 10	N = 30	N = 100
seed / final feasible cost	8 / 8.445889	generations in 100 000 evals	11110	3447	1009
improving transfers / swaps	0 / 20	mean cost (gap_ref %)	8.4078 (0.30)	8.4143 (0.38)	8.4862 (1.23)
distinct individuals, first / last generation	30 / 30	worst cost (gap_ref %)	8.4235 (0.49)	8.4459 (0.75)	8.5501 (2.00)
checked moves (all / feasible)	500 / 353 transfers; 1114 / 667 swaps	std; feasible runs	0.0119; 10/10	0.0184; 10/10	0.0334; 10/10

Improving = feasible and Δ < −1e-09; first generation = initial population, last = last complete generation (index 3447). N = 30 is the rule value and stays in Table 1; the N runs are a comparison on seeds 0–9, not a tuning.

3. The six questions

Q1 Feasible? Yes: 10 of 10 runs (and 10 of 10 for every N and for random search) pass the checker; there is no infeasible run, so no overload to report. The repair keeps every individual within CPU, memory and zone, and the instance is far from full (46 % / 50 %).

Q2 Repeatable? Only partly. Best 8.3828 (seed 4, equal to z_{ref} up to 1e-10), median 8.4123, worst 8.4459 (seed 8), std 0.0184. The spread worst − best is 0.063 = 0.75 % of z_{ref}, twice the mean gap (0.38 %); 7 of 10 runs are within 0.5 % of z_{ref}. Different seeds end in different local optima.

Q3 Converged? The mean curve is within 1 % of its final value after 11 % of the budget, within 0.5 % after 26 % and within 0.2 % after 58 %. In the last 10 % of the budget the mean improves by only 0.0015 (0.02 %) and only 1 of 10 runs improve at all: the curve is flat at the end (stuck), not budget-limited. Yet N = 10 (more generations) is better, so the budget is not irrelevant.

Q4 Local optimum of which neighbourhood? Worst seed 8, final cost 8.44589 (gap_ref 0.75 %): of 500 transfers (353 feasible) **0 improve**, the least bad one costs Δ = +0.0031 (VM 29, server 5 → 10); of 1114 swaps (667 feasible) **20 improve**, the best one (VMs 6 and 14, servers 10 ↔ 5) gives Δ = -0.0309, which alone would bring the run to 8.4150, close to the median. So the final solution is a local optimum for transfers but not for swaps: a mutation of 1-2 components can make this swap only by hitting two given VMs *and* their two given servers, and the repair never proposes moves of this kind. The run reached its final value at evaluation 85 500 (last 14.5 % of the budget without improvement); the CPU utilisation of the servers is uneven (0.25-0.59).

Q5 What did the method do? Distinct individuals: 30 in the first and 30 in the last generation of the worst seed (mean over seeds 30 and 30, minimum over all generations and seeds 29): no identical individuals, no collapse. The mean pairwise Hamming distance falls from 44.0 (random start) to 16.1 in generation 100 and then stays at 15.5 of 50 until the end (F2): a balance of mutation, crossover and selection, not a convergence. Two parents that differ in ~15 components give by uniform crossover a child that differs from each parent in about 7-8 components, so crossover keeps making large, mostly worse steps (my explanation, consistent with the numbers; I did not test it).

Q6 Better than random search? Yes, on all 10 seeds: mean gap$_{ref}$ 0.38 % against 5.53 %, mean gap$_{LP}$ 0.89 % against 5.73 % (random search used the same repaired random individuals and the same budget). The relaxation itself is 0.52 % below z_{ref} : of the 0.89 % mean gap to z_{LP} , 0.52 points are the gap of the relaxation (the best run, which equals z_{ref}, shows exactly this) and only about 0.38 points are the GA's own gap.

4. Required item: diversity and population size; prediction against result

Population size. With the same budget N = 10 gives 11110 generations, N = 30 3447 and N = 100 only 1009. N = 100 is clearly worse (mean 8.4862 against 8.4143; difference 0.072, standard error ≈ 0.012, about 6 standard errors; it also has the larger std 0.0334): a big population leaves only 1009 generations for the same budget, and each generation improves the best individual only slowly (I did not plot the N = 100 curves). N = 10 is slightly better than N = 30 (mean 8.4078, worst 8.4235), but the difference 0.0065 is about 0.9 standard errors (0.0069), so with 10 seeds I cannot call it a real gain. More individuals do not help here; more generations do.

Prediction against result. Feasibility: right (10/10). Mean gap$_{ref}$ 0.38 % lies in the predicted 0.2-1.5 % (the worst run, 0.75 %, as well). Distinct individuals: right in the sense that they do not collapse - but they stayed at 30 of 30, much higher than my guess "above ~10". **Wrong:** I expected the Hamming distance below ~5 within 10-20 % of the budget; it fell to ~16 within about 3 % of the budget (generation 100) and stayed there, so the population did not converge prematurely; the stalled runs are therefore not a loss of diversity - my reading (not tested separately) is a crossover that is too disruptive plus the missing swap-like step (Q4). "A larger N will not help" was right for N = 100 (worse), not for N = 10 (equal or slightly better).